Q.1. From the perspective of maximizing likelihood of parameters, derive the gradient descent/ascent update rules using the appropriate cost functions for the following:

i) Linear Regression

ii) Logistic Regression

Q.2. [P] For the provided dataset, implement a linear regression model to predict housing prices. Train the model using both the gradient descent and the normal equations method. Show the performance of your model on the test set using a metric of your choice.

Q.3. [P] The following data shows the force on a projectile due to air resistance at different velocities in a wind tunnel experiment. You are required to estimate the force when the velocity of the projectile is 750 ft/sec. You are required to do the following using any linear algebra library, e.g. NumPy.

i) Find an interpolating polynomial $p(x) = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5$

ii) Fit a quadratic polynomial $p(x) = c_0 + c_1x + c_2x^2$ using regression.

Velocity (100 ft/sec)	0	2	4	6	8	10
Force (100 lb)	0	2.90	14.8	39.6	74.3	119

What is the difference between the approaches (fitting vs interpolating) mentioned above?

Q.4. [P] For the provided dataset, implement a classifier using logistic regression as follows:

i) Using a linear decision boundary

ii) Using a non-linear decision boundary

Give a comparative analysis of the performance of the above classification models.

Submission Notes:

1. The deadline for submission is 11:59 PM on September 30, 2026
2. Submit a soft copy along with code through email: ali.ds@pu.edu.pk

[P] after a question means it's a programming exercise. Use Python or any language that you prefer.